

PRD v3.0 (Final) — Dr. Siddharth Gupta Digital Healthcare Platform

Oct 6, 2026 · @RINTU · Final for build · Companion: Build Plan — Web, iOS and Android

1. Document control

This is the single, consolidated product requirements document for build. It replaces PRD v1, v2, v2.1 and v2.2; nothing in earlier versions applies unless repeated here. How it is built (architecture, platforms, team, timeline) is in the companion Build Plan.

Field	Value
Product	Dr. Siddharth Gupta Digital Healthcare Platform
Version	3.0 — Final for build
Demo	Approved (clickable prototype, mock data)
Launch customer	Dr. Siddharth Gupta, Aceso Clinic (orthopaedics assumed, to confirm)
Platforms	Web; Patient app and Clinic app on iPhone, iPad and Android
Target	Clinic pilot about 7 months after team start; store launch about 1 month later

1.1 Key product decisions

Area	Decision
Product shape	Multi-doctor, multi-clinic platform; launches with one doctor
Apps	Web app plus two mobile apps (Patient, Clinic) from one codebase
Doctor's main surface	Clinic app on phone and iPad; full web dashboard as well
Front desk's main surface	Web dashboard; Clinic app on tablet as backup

Area	Decision
Prescription signing	Cloud Class 3 DSC, approved by doctor PIN and biometrics
Payments	Cashfree Payments: links, in-app and web checkout, refunds; each clinic uses its own Cashfree account
Messaging	WhatsApp first, SMS fallback and OTP, email for documents, push for app users
AI	Voice scribe that drafts notes; read-only Ask AI over one patient's record; never diagnoses or signs
ABDM	ABHA, Scan & Share and record sharing (HIP) at launch; fetching records (HIU) within 6 weeks
Hosting	India region (AWS Mumbai)

1.2 Open items to close before build starts

- ☐ Confirm specialty (orthopaedics) and the share of child patients.
- ☐ Legal entity that owns app store accounts, cloud, domain and code.
- ☐ Vendors: cloud DSC, drug database, AI scribe (speech and model), WhatsApp provider, SMS provider.
- ☐ Cashfree merchant KYC and settlement account for the clinic; agreed fees; whether pay-at-booking is required for in-person visits.
- ☐ Online consultation tool for V1 (phone call or external video link).
- ☐ Follow-up fee rule and fee list.
- ☐ Multi-doctor chart model default (shared chart or own patients only).
- ☐ Data export available from the current EMR.

2. Vision, goals and scope

A patient connects with Dr. Gupta digitally — books, pays, consults, receives a legally signed prescription and keeps all records in one place — whether they use the app, a browser, or only walk in. The doctor runs his whole OPD from his phone or iPad faster than on his current EMR, and the platform can onboard other doctors later without a rebuild.

2.1 Goals

1. Patients book, pay, reschedule and cancel without a phone call.

2. App bookings, phone bookings, walk-ins and Scan & Share arrivals land in one live queue.
3. The doctor sees full context (history, allergies, vitals, reports, ABDM records) on one screen.
4. A typical prescription takes under 60 seconds, helped by the AI scribe, templates and repeat-last.
5. Every prescription passes the safety engine, is DSC-signed, immutable, and reaches the patient by WhatsApp, SMS, app or print.
6. Bills are paid by payment link or in-app checkout and reconciled automatically.
7. Reminders and follow-up booking reduce no-shows; review links grow the practice.
8. Nothing the doctor uses daily on his current EMR is missing on day one.

2.2 Non-goals

- Hospital management (beds, OT, inpatient, pharmacy inventory). This is an outpatient platform.
- In-app video calling (V1 uses a phone call or an external video link).
- AI that diagnoses or prescribes; the AI drafts and answers, the doctor decides and signs.
- OCR of uploaded reports; partner lab and pharmacy ordering; insurance claims.
- Self-serve doctor sign-up and subscription billing (onboarding is led by the platform team).
- Accounting ledger or inventory.

2.3 Assumptions

- One organisation, one doctor, one clinic at launch; more of each without code changes.
- Patients in India; all times IST; English and Hindi.
- Most patients prefer WhatsApp or SMS over email, and many will never install the app — every patient output must also work by link.
- About 50–100 appointments a day at launch.

3. Users, roles and platforms

Every staff user belongs to one organisation and sees only its data. Clinical content is visible only to treating doctors and to the patient it belongs to.

3.1 Roles and where they work

Role	Who	Main surface	Also available on
Patient	Account holder (phone number) managing self and family	Patient app (iPhone, Android)	Patient web portal, iPad
Doctor	Dr. Gupta at launch; onboarded doctors later	Clinic app (phone, iPad, Android tablet)	Web dashboard
Front desk	Clinic staff assigned to doctors	Web dashboard	Clinic app on tablet or phone (daily tasks)
Clinic Admin	Practice manager	Web	—
Platform Admin	Platform operator's team; no clinical access	Web platform console	—
Verifier	Pharmacist or anyone scanning the prescription QR	Public verify page	—

3.2 Permissions

Capability	Patient	Front desk	Doctor	Clinic Admin
Manage own profile and family	Yes	—	—	—
Register patients, walk-ins, Scan & Share	—	Yes	Yes	Yes
Book, reschedule, cancel	Own	Any	Any	Any
Check in, no-show, queue, tokens	—	Yes	Yes	—
Bills, payment links, counter payments	Pay own	Yes	Yes	View, refunds
Vitals	View own	Enter	Yes	—
History, notes, diagnosis	Own summary	—	Yes	—

Capability	Patient	Front desk	Doctor	Clinic Admin
Uploaded documents	Own	Upload on behalf	Yes	—
Prescriptions: create, sign, amend	—	—	Yes	—
Prescriptions: view, print	Own	Print	Yes	—
Tags and UHID	—	Assign	Assign	Configure
Availability and fees	—	If delegated	Yes	Yes
Templates, medicine master	—	—	Templates	Master
Settings, messaging templates, price list, Cashfree setup	—	—	—	Yes
Audit log, data-rights requests	—	—	Own actions	Yes

Platform Admin manages organisations, doctor verification, usage and system health only — no scheduling, billing or clinical rights.

Chart model per organisation: *shared chart* (every doctor in the clinic sees full history) or *own patients only* (access after treating or receiving a referral).

3.3 Sign-in

- Patients: mobile OTP by SMS (WhatsApp OTP optional); biometric unlock on device.
- Doctors and front desk: password + authenticator code on new devices; Face ID or fingerprint on registered devices; app locks after 15 minutes idle; lost devices revoked from web.
- Admins: password + authenticator, web only; platform console restricted by IP.

4. Patient experience

One phone number is one account; one account holds several patients (family). Every patient feature works in the app and in the web portal, so a patient sent a link never has to install anything.

4.1 Onboarding and profile

- Language (English/Hindi) → phone → OTP (6 digits, 5-minute validity, limited retries) → privacy notice, terms and separate health-data consent → name, date of birth, gender → home.
- If the clinic already registered the number, the patient claims that record instead of creating a duplicate.
- Profile: contact details, emergency contact, photo; medical basics (blood group, allergies, conditions, current medicines) marked "pending doctor verification".
- Family members with their own records, UHID and ABHA; under-18s linked to a guardian.
- Change phone number with OTP on old and new numbers; lost old number → in-person verification at the clinic.
- Channel preferences (WhatsApp, SMS, email, push); reminders can be muted, essentials cannot.
- Data rights: download my data, request correction, delete account (clinical records retained for the legal period).

4.2 Home

Patient switcher, next appointment (date, time, mode, token, join or directions, countdown), quick actions (Book, Appointments, Records, Prescriptions, Pay), follow-up reminder, latest prescription, health snapshot (last BP, weight, SpO₂), messages and notifications.

4.3 Booking

1. Choose patient → doctor (when the clinic has several) → mode (in-person, phone, video) → date → slot.
2. Reason for visit; attach reports.
3. Online modes: identity check and telemedicine consent.
4. Review fee, follow-up eligibility and cancellation policy.
5. Pay with Cashfree if required (always for online consultations; optional per clinic for in-person). Slot held for a few minutes while paying.
6. Confirmation in app and on WhatsApp/SMS, with an add-to-calendar option.

Booking rules (configurable): 30-day horizon, 60-minute same-day cutoff, auto-confirm or manual confirm, reschedule/cancel up to 2 hours before, maximum 2 active bookings per patient, no-show limit (e.g. 3 in 90 days blocks self-booking), follow-up window and fee.

4.4 Appointments, records and prescriptions

- Appointments: upcoming and past, status timeline, reschedule and cancel within rules, visit summary in plain English or Hindi.
- Records: categories (prescriptions, lab, imaging, discharge, other); upload from camera, gallery or files (multi-file, multi-page photos combined into one PDF); link to an appointment or a test order; "Reviewed by doctor" badge; delete unless the doctor has used the document; share by time-limited link.
- Prescriptions: list, PDF viewer, download, share; amendments shown as "Revised on ..." with earlier versions viewable; voided ones watermarked.
- Test orders: see what was ordered and upload results against it.
- Assessments: fill forms sent by the clinic before a visit.

4.5 Payments, messages, ABHA and consents

- Pay dues and open payment links; receipts saved in the app.
- Messages view mirrors WhatsApp messages, with working buttons (Confirm, Reschedule, Cancel, Book follow-up, Pay).
- ABHA: create or link for self and family; ABHA card with QR.
- Consent manager: approve, deny or revoke ABDM record-sharing requests and AI scribe consent requests.

5. Clinic operations

One queue serves the doctor and the front desk, on every device, updating live.

5.1 Queue

- Tabs: **My OPD** (checked in, waiting, in consultation), **Booked** (not arrived; unconfirmed show "Awaiting response"), **Completed**, **Follow-ups** (patients due back).
- Date selector, search, filters by tag, doctor, mode and payment status; separate or combined queues per doctor.
- Card: token, booked time, live waiting time, walk-in or booked icon, name, age, gender, tags, billed amount.
- Card actions: Write Rx, Scribe, Exit, Check in, Reschedule, Bill, View Rx, Repeat visit.
- Card sheet: call patient, ABHA, tags, create receipt, send payment link, add vitals, add records, assessment, refer, send review link, start video call, cancel.
- Visit duration recorded automatically; average consultation time on the dashboard.

5.2 Appointment statuses

Pending → Confirmed → Checked-in → In consultation → Completed; plus Cancelled (reason), No-show (automatic after the slot, or manual) and Rescheduled (linked to the new appointment). Every change stored with who and when.

5.3 Front desk

- Walk-in: search by phone or UHID → existing patient or quick register → token.
- Scan & Share: patient scans the clinic's ABDM QR and arrives in the queue with details prefilled.
- Phone bookings on a patient's behalf; overbooking within the clinic's limit.
- Check-in, vitals, counter payments, printing (prescription, receipt, day sheet).
- Doctor running late: broadcast the delay; doctor unavailable: cancel or bulk-reschedule a session with notifications.
- WhatsApp inbox for patient replies (24-hour reply window shown).
- Duplicate patient detection and merge requests; lost-phone verification and number change.
- Waiting-room display: current and next tokens, no names.

5.4 Tags and UHID

- Tags configurable per organisation with colours (defaults: Priority, Emergency, 2nd opinion, VIP, Insurance, Complaint); Emergency and Priority can sort to the top of the queue.
- UHID with clinic prefix and starting number (e.g. EK10001); imported patients keep their existing UHID; shown on profile, queue, prescriptions and receipts.

5.5 Availability

Weekly schedule per consultation mode (days, sessions, slot length, buffer, breaks); leave and holidays with impact on existing bookings; extra sessions; schedule versions with an effective-from date so changes never disturb booked days.

5.6 Billing and payments (Cashfree)

Cashfree Payments is the gateway for all online collection. Each organisation connects its own Cashfree merchant account, so money settles directly to the clinic.

Flow	How it works
Payment link	Bill → Send payment link → Cashfree link sent on WhatsApp (SMS fallback) → patient pays by UPI, card, net banking or wallet → verified webhook marks Paid → receipt sent
Pay at booking	Cashfree order created at booking; Patient app uses Cashfree's SDK (UPI apps open directly), web uses Cashfree checkout; slot held until confirmed or expired
Pay dues	Outstanding bills payable from app and portal
Counter payment	Cash, clinic UPI QR or card machine recorded by front desk
Refund	Full or partial, with reason, through Cashfree; patient notified

- Bill: consultation fee plus price-list items (plaster, dressing, injection, X-ray review), discount, total. Statuses: Billed, Link sent, Paid, Partly paid, Due, Refund pending, Refunded.
- The server sets every amount; a payment is successful only after a verified Cashfree webhook or server status check; duplicate webhooks are ignored.
- Payment links expire after 48 hours by default; resending cancels the old link.
- Card and UPI details are handled only by Cashfree.
- Consultation fees are a real-world service, so app-store in-app purchase does not apply.
- Daily reconciliation against Cashfree settlements: gross, fees, net, refunds, dues; mismatches flagged; CSV export for the accountant.
- Collection reports by mode and doctor; receipts numbered per organisation (tax treatment confirmed with the clinic's CA).

6. Consultation and prescribing

Speed comes from the scribe, templates and repeat-last; safety from the safety engine; legal validity from DSC signing and immutability.

6.1 Consultation screen

- Context (left on tablet/web, top on phone): name, age, gender, UHID, allergies highlighted, conditions, current medicines, recent visits, last prescription, unreviewed reports, ABDM records, tags.
- This visit: vitals (BP, pulse, temperature, SpO₂, weight, height, BMI, pain score; pregnancy/breastfeeding where relevant; weight required under 12), chief complaint, symptoms with duration, examination, diagnosis with ICD-10 search and favourites, plan, private notes, tests advised, advice, follow-up date with one-tap booking.
- Autosave; locks once the prescription is signed.

6.2 Orthopaedic pack (to confirm)

Examination template (site, side, swelling, tenderness, range of movement, deformity, neurovascular status); diagnosis favourites (sprains, ligament injuries, fractures, knee osteoarthritis, low back pain, frozen shoulder); imaging orders by region and side; advice library (ice compression/RICE, physiotherapy sheets, brace/splint, weight-bearing); billable procedures (plaster, dressing, joint injection); ortho prescription templates. Body diagram in V1.1.

6.3 AI voice scribe

- Modes: ambient (doctor and patient) and dictation (doctor alone). Languages: English, Hindi, Hinglish.
- Ambient recording starts only after patient (or guardian) consent, captured on screen or on the patient's phone.
- Live transcript with speaker labels; on stop, a structured draft fills complaint, history, examination, assessment, plan, tests, advice and follow-up, and suggests medicines.
- AI-filled fields are marked Draft; the doctor accepts, edits or discards each section; signing is blocked while drafts are unaccepted; suggested medicines go through the safety engine.
- Draft within 30 seconds for a 10-minute consultation; works through short network drops; doctor can always type instead.
- Audio deleted after the retention window (default 7 days); transcript kept with the visit; model version recorded.
- Patient summary generated only from accepted text.

6.4 Prescription builder

- Medicine search across brand and generic, showing strength, form and safety tags; free-text medicine allowed and flagged for the master.

- Line: dose, frequency (1-0-1 style or free text), timing, duration, route, quantity, instructions.
- **Auto-written remarks** in English or Hindi from dose, frequency, timing and duration — e.g. "Take 5 ml three times a day, after breakfast, lunch and dinner, for 5 days"; supports SOS, weekly/monthly schedules and tapering; editable.
- Templates, repeat last prescription, save as template.

6.5 Safety engine

Checks run as each line is added and again before signing, against the patient's chart and a licensed drug database with Indian brand mapping.

Check	Severity
Allergy: direct or same class	Block
Allergy cross-sensitivity (e.g. penicillin → cephalosporin)	Warn
Drug interaction: contraindicated / major / moderate-minor	Block / Warn with reason / Info
Duplicate molecule or class	Warn
Drug-condition (e.g. NSAID with kidney disease)	Warn with reason
Pregnancy, breastfeeding: contraindicated / caution	Block / Warn
Child dosing: no weight today, or dose outside mg/kg range	Block
Maximum daily dose exceeded	Block
Older adults (65+) caution	Info
Kidney or liver adjustment	Warn
Telemedicine drug lists: Prohibited online / List B outside follow-up	Block / Warn
Missing dose, frequency or duration	Block

Blocks stop signing (override only where the database allows, with a reason); warnings need acknowledgement; every alert and action is logged; allergy and contraindicated blocks cannot be switched off; patient-reported data is checked but labelled unverified.

6.6 Signing, immutability and delivery

- Preview → Sign (biometric or PIN) → server re-checks → PDF generated → cloud Class 3 DSC signs → locked → delivered. Only verified doctors can sign.
- Signed prescriptions are immutable. Corrections create a new version with a reason; voiding watermarks the PDF; the patient is notified each time.
- PDF (A4/A5, prints in black and white): doctor name, qualifications, registration number and council, clinic details; patient name, age, gender, UHID, date, prescription number with the doctor's prefix, consultation mode; vitals, symptoms, diagnosis, allergies; medicines table (brand plus composition, generic in capitals, dose, frequency, duration, remarks); prescribed tests; advice; follow-up; signature block; verification QR; disclaimer.
- Delivery: app, WhatsApp, SMS link (opens after OTP or date-of-birth check), email copy, print at the desk, ABDM for ABHA-linked patients. Pharmacies verify by QR.

7. Orders, referrals, assessments and Ask AI

These close the loop around the consultation: what the doctor asks for, who else is involved, and what the record already says.

7.1 Test orders

- Catalogue of blood tests, X-ray (with views), MRI and CT (with region and side, e.g. "MRI knee, right").
- Orders print under "Prescribed tests" and appear in the Patient app with Upload result.
- Uploaded results link to the order; the doctor is notified; status Ordered → Result uploaded → Reviewed.
- Partner lab booking: Phase 2.

7.2 Referrals

- Refer inside the organisation (pick a doctor) or outside (name, specialty, hospital, phone) with reason, urgency, notes and attachments.
- Signed referral letter PDF sent to the patient and the receiving doctor.
- Internal referral grants chart access under the "own patients only" model.
- Status Sent → Seen → Closed, listed on the doctor dashboard.

7.3 Attachments

Send any document to a patient (exercise sheet, diet chart, report) on WhatsApp and in the app; clinic library of reusable sheets.

7.4 Health assessments

- Form builder with scored questions (scale, choice, yes/no, number). Third-party validated scales only where their licence allows; otherwise the clinic's own wording.
- Sent by WhatsApp link before a visit or started in clinic; results stored with score and a trend over time; Start assessment and Past assessments on the profile.

7.5 Ask AI about this patient

- Chat panel on the patient profile answering only from that patient's record (visits, prescriptions, reports, ABDM records) — e.g. "When was the last MRI?", "Which painkillers in the last year?", "Summarise the last three visits".
- Every answer shows the record entries it used, with links; if the answer is not in the record, it says so.
- Read-only: never diagnoses, prescribes or writes to the chart. Same consent, vendor and audit rules as the scribe; every question logged.

7.6 Google review link

- Per-doctor Google Business Profile link.
- Optional automatic request after a completed visit (default 2 hours later), at most once per patient every 90 days, never after a cancelled visit or for patients tagged Complaint; manual send from the card sheet.
- Sent as an approved WhatsApp marketing template with SMS fallback; sent and clicked counts per doctor.

8. Messaging and ABDM

WhatsApp is the primary patient channel, SMS the guaranteed fallback and OTP channel, email carries documents, and push serves app users.

8.1 Messages by event

Event	WhatsApp	SMS	Email	Push
OTP	Optional	Primary	—	—
Booking confirmed / rescheduled / cancelled	Primary	Fallback	With calendar file	Yes
Reminder 24 h and 2 h before (Confirm / Reschedule / Cancel buttons)	Primary	Fallback	—	Yes

Event	WhatsApp	SMS	Email	Push
Doctor late / session cancelled	Primary	Always	—	Yes
Online consult join details	Primary	Always	—	Yes
Payment link / receipt	Primary	Fallback	Receipt PDF	Yes
Prescription available	Primary (secure link)	Fallback	PDF copy	Yes
Prescription amended / voided	Primary	Always	Yes	Yes
Follow-up due (Book button)	Primary	Fallback	—	Yes
Assessment or test-result request	Primary	Fallback	—	Yes
ABDM consent request / record shared; scribe consent request	Primary	Fallback	—	Yes
Review request	Primary (marketing template)	Fallback	—	—
Account and security	—	Always	Yes	Yes

8.2 Messaging rules

- If WhatsApp is not delivered within 5 minutes, or the patient has no WhatsApp or opt-in, the message goes by SMS. "Always" events go on both.
- Opt-in recorded at sign-up, booking or by the front desk; STOP honoured immediately.
- WhatsApp buttons update appointments and open payment directly; free-text replies land in the front-desk inbox.
- SMS templates DLT-registered; WhatsApp templates Meta-approved; English and Hindi; managed per organisation.
- Each organisation sends under its own WhatsApp number, SMS sender ID and email domain.
- Messages never contain diagnoses or medicine names; they link to the app or an authenticated page.
- Every send logged with channel, status, retries and cost.

8.3 ABHA and ABDM

Milestone	What it does	Release
M1 ABHA	Create or link ABHA for patients and family members	Launch
Scan & Share	Patient scans the clinic QR; arrives in the queue with details	Launch
M2 Health Information Provider	Consultations and prescriptions linked as care contexts; shared as FHIR records with patient consent	Launch
M3 Health Information User	Doctor requests records from other providers with patient consent (purpose, types, dates, expiry); shown read-only with source	Within 6 weeks of launch

Facility registered on HFR, each doctor on HPR; sandbox integration and NHA review before production; ABHA optional for patients; consents stored and revocable.

9. Multi-doctor platform, administration and website

The platform is multi-tenant from day one: organisation (practice) → clinics (locations) → doctors and staff. New doctors are added by configuration, not development.

9.1 Doctor onboarding (Platform Admin)

1. Organisation and clinic.
2. Invite doctor; doctor sets password and authenticator.
3. Registration number, council, qualifications, specialty, HPR ID, certificate upload.
4. Platform Admin verifies (Pending → Verified → Active); unverified doctors cannot sign.
5. DSC setup and test signature.
6. Availability, consultation modes and fees.
7. Prescription pad: logo, header, footer, language, specialty templates, prescription number prefix.
8. Branding: display name, colours, profile page, booking link, WhatsApp sender, SMS sender ID, email domain, Google review link.
9. Cashfree merchant account connected (or organisation's existing one) and tested.
10. Go-live checklist → Active.

9.2 Clinic Admin

- Dashboard: patients, appointments, completed, cancellations, no-show rate, prescriptions, collections, message cost.
- Patients and appointments (no clinical content), merges, data-rights requests.
- Settings: booking rules, fees and price list, tags, UHID, chart model, safety tuning (non-critical only), scribe defaults, file limits.
- Messaging templates, sender settings, delivery log.
- Medicine master and approval queue for free-text medicines.
- Cashfree: account status, payment methods, link expiry, refunds, reconciliation reports.
- Staff accounts, roles, device revocation; audit log.
- Import tool for migration.

9.3 Platform console

Organisations and doctors with status, verification queue, usage per organisation (appointments, prescriptions, messages, scribe minutes, payments), system health. No clinical data.

9.4 Public website

- Organisation site and a profile page per doctor (qualifications, registration, services, timings, map, FAQs), each with its own booking link and an embedded booking widget with Cashfree checkout.
- SEO metadata, doctor schema markup, sitemap; app store badges.
- Prescription verification page (QR target) showing Genuine, Superseded or Void.

9.5 Migration from the current EMR

1. Request a full export (patients with UHIDs, visits, prescriptions, receipts, attachments).
2. Import wizard: upload, column mapping, duplicate detection, dry-run report, import; imported prescriptions shown as historical, read-only.
3. Fallback: patient list plus old PDFs; ABDM fetch for ABHA-linked patients.
4. Two-week parallel run, fixed cut-over date, old system kept read-only for the retention period.

10. Compliance and privacy

Compliance items are launch blockers. They state what the product must support and must be confirmed by a healthcare lawyer before build sign-off; this is not legal advice.

ID	Source	Product requirement
C1	Telemedicine Practice Guidelines 2020	Verify patient identity before online consults; record guardian for minors
C2	Telemedicine Practice Guidelines 2020	Telemedicine consent (implied when patient books; explicit when doctor initiates), stored with time
C3	Telemedicine Practice Guidelines 2020	Registration number and council on every prescription, profile and website
C4	Telemedicine Practice Guidelines 2020	Medicines tagged List O/A/B/Prohibited; Prohibited blocked online; List B warned outside follow-up
C5	Telemedicine Practice Guidelines 2020	Consultation mode recorded on every consultation and prescription
C6	IT Act 2000	Legally recognised electronic signature (cloud Class 3 DSC)
C7	IMC Regulations 2002	Consultation and prescription records retained at least 3 years; not patient-deletable
C8	IMC Regulations / NMC guidance	Medicine names legible; generic names in capitals on the PDF
C9	DPDP Act 2023 and Rules	Privacy notice, purpose-specific consent, withdrawal, access/correction/erasure requests, grievance contact, breach process
C10	DPDP Act 2023	Verifiable guardian consent for patients under 18
C11	TRAI DLT	Entity, sender ID and SMS templates registered before go-live
C12	Apple and Google	Health declarations, privacy labels, data safety form, in-app account deletion, organisation accounts

ID	Source	Product requirement
C13	ABDM (NHA)	HFR/HPR registration, sandbox, NHA review, India-hosted data, consent artefacts
C14	DPDP Act; Telemedicine Guidelines	Explicit per-consultation consent before ambient scribe recording
C15	DPDP Act	AI vendors under data processing agreements, no training on our data, approved regions, audio auto-deleted
C16	Meta WhatsApp policy	Opt-in before WhatsApp, approved templates, immediate opt-out
C17	Medical liability	AI text marked draft until accepted; safety overrides need a reason and are logged
C18	RBI rules for payment aggregators (via Cashfree)	Payments only through Cashfree; no card or UPI credentials stored; refund policy published on website and apps

10.1 Retention and erasure

- Doctor-created records (consultations, prescriptions, vitals, notes, signed PDFs, safety logs): retained for the legal period; never patient-deletable; erasure requests honoured after retention ends.
- Patient uploads: deletable unless the doctor used them, in which case hidden from the patient but retained.
- Account deletion: login and profile deactivated immediately; clinical records follow the rule above.
- Scribe audio: deleted after the retention window; transcripts retained with the visit.
- Payment records: retained as required for financial records.

10.2 Documents required at launch

Terms of Use, Privacy Policy, telemedicine consent, AI scribe consent, refund and cancellation policy, data processing agreements with every vendor (hosting, SMS, WhatsApp, email, Cashfree, DSC, drug database, AI).

11. Non-functional requirements and data model

Targets are sized for launch with one doctor and scale horizontally to tens of doctors. Architecture details are in the Build Plan.

11.1 Performance and reliability

Requirement	Target
Uptime during clinic hours	99.5% monthly
API response (p95)	Under 400 ms
Prescription PDF	Under 5 s
Scribe draft (10-minute consult)	Under 30 s
Payment confirmation after Cashfree webhook	Under 1 minute
Patient app cold start (mid-range Android)	Under 3 s
Backups	Daily full + continuous logs; recovery point 15 min, recovery time 4 h; monthly restore test
Network drops	Today's queue, opened summaries and drafts work offline; writes sync later

11.2 Security

- India-region hosting; TLS everywhere; encryption at rest; field-level encryption for notes, diagnoses, transcripts and signing material.
- Organisation isolation enforced in the database and API, tested on every change.
- Private file storage with short-lived signed links; malware scan on upload.
- Role checks on every request; rate limits on OTP, login, upload and payment endpoints.
- Signed and de-duplicated webhooks (Cashfree, WhatsApp, SMS, ABDM).
- Append-only audit log for record views, prescription sign/amend/void, safety overrides, document access, payments and refunds, permission changes, exports and logins.
- Secrets in a managed store; Cashfree keys per organisation, encrypted.
- Penetration test before pilot and yearly; breach response runbook with a named owner.

11.3 Compatibility and accessibility

iOS/iPadOS 16+, Android 9+ (target the API level Play requires), latest two versions of major browsers; usable on low-end Android and slow networks; WCAG 2.1 AA on web, dynamic text sizes on mobile; English and Hindi.

11.4 Data model

All tables carry an organisation ID, created/updated times and soft delete where needed.

Group	Entities
Platform and identity	Organisation, Branding, Clinic, User, Session, Device, StaffMember, Doctor, DoctorClinic, DoctorOnboarding
Patients	Patient, Tag, PatientTag, Allergy, MedicalCondition, CurrentMedication, Consent, MessagingOptIn, AbhaLink
Scheduling	ConsultationType, AvailabilityVersion, AvailabilityException, Appointment, AppointmentStatusHistory, VisitTiming
Clinical	Consultation, Vitals, ScribeSession, Assessment, AssessmentForm, AskAiQuery
Prescribing	Medicine, DrugMolecule, PrescriptionTemplate, Prescription, PrescriptionItem, PrescriptionTest, SafetyAlert
Orders and referrals	TestOrder, TestResult, Referral, Attachment, MedicalDocument
Billing and payments	PriceListItem, Bill, BillItem, Payment, PaymentGatewayAccount (Cashfree), PaymentLink, Refund, Settlement, Receipt
Messaging	MessageTemplate, Notification, MessageThread, DeliveryLog, WebhookEvent
ABDM	CareContext, HealthConsent, ExternalRecord
Governance	AuditLog, DataRightsRequest, ImportJob

Files are stored by key, never by public URL. Prescriptions store PDF hash, template version, signature method and version chain. Payments store Cashfree order, payment and link IDs, method, fee and settlement date.

12. Release scope, acceptance and launch

Launch happens when the full journey — book, pay, check in, consult with or without the scribe, sign, deliver, bill, follow up — works on web, iPhone, iPad and Android for app users, link users and walk-ins, with C1-C18 in place.

12.1 Scope by release

Capability	Launch	Shortly after launch / Phase 2	Future
Web app, Patient app, Clinic app (iPhone, iPad, Android)	Yes	White-label Patient apps per organisation	
Booking, queue tabs, front desk, tags, UHID, visit timing	Yes	Waitlist	
Consultation, ortho pack, AI scribe (English, Hindi, Hinglish)	Yes	More languages, body diagram	AI summaries across visits
Safety engine, DSC signing, versioned PDF, QR verification, dosage remarks	Yes		Lab-value checks
Test orders, referrals, attachments, assessments, Ask AI, review links	Yes	Partner lab booking	
Cashfree payments: links, in-app and web checkout, refunds, reconciliation	Yes	Platform fee via Cashfree split payments	Subscriptions
WhatsApp, SMS, email, push with fallback and inbox	Yes		
ABDM M1, Scan & Share, M2	Yes	M3 within 6 weeks	
Multi-doctor platform, onboarding, platform console	Yes (one doctor live)	Self-serve doctor sign-up	Multiple clinics per doctor at scale

Capability	Launch	Shortly after launch / Phase 2	Future
Migration from current EMR	Before go-live		
In-app video, pharmacy, insurance, OCR		Phase 2	Future

12.2 Acceptance criteria (must pass on staging before pilot)

- Two patients booking the same slot at once: exactly one succeeds.
- Penicillin allergy blocks amoxicillin and warns on a cephalosporin; a contraindicated interaction blocks signing; a major interaction needs a reason that appears in the audit log.
- A child under 12 with no weight today cannot get a weight-based drug.
- A Prohibited-list medicine cannot be prescribed in an online consultation.
- Signed prescriptions cannot be edited; an amendment creates version 2 and notifies the patient; the QR shows Genuine, Superseded or Void correctly.
- Ambient scribe cannot record without consent; draft within 30 seconds; nothing becomes the note until accepted.
- A Cashfree payment link paid by UPI marks the bill Paid and sends a receipt within a minute of the webhook; a duplicate webhook records one payment; a forged webhook changes nothing; closing the app after paying still ends Paid.
- An online consultation cannot be confirmed without payment; an abandoned payment releases the slot.
- A patient without WhatsApp receives the reminder by SMS within 5 minutes; tapping Reschedule on WhatsApp opens the slot picker.
- Scan & Share puts the patient in the queue with ABHA details; a prescription is retrievable through ABDM after consent.
- No user of organisation A can read organisation B's data through any API, search or file link.
- A new doctor is onboarded end to end in under an hour without code changes.
- The doctor completes a full clinic day on iPad, on a phone and on web.

12.3 Success metrics (3 months after launch)

Metric	Target
Appointments booked digitally	50% or more
No-show rate	30% lower than baseline
Prescriptions issued digitally	90% or more
Median time to complete a prescription	Under 60 s
Consultations using the scribe	50% or more
Bills paid online through Cashfree	40% or more
Follow-ups booked from the consultation	40% of advised follow-ups
Patients with linked ABHA	25% or more

Baselines measured for 2–4 weeks before launch.

12.4 Key risks

Risk	Mitigation
ABDM or app-store review delays	Start week 1; ABDM behind a feature flag; organisation accounts and clear review notes
Scribe mishears a drug or dose	Draft-only, highlighted, safety-checked, accepted section by section
Alert fatigue	Severity tiers; only blocks interrupt; non-critical tuning
Cashfree activation or KYC delay	Start KYC in week 1; counter payments work regardless
Cross-organisation data leak	Database-level isolation, tests on every change, focused pen test
Incomplete data from current EMR	Early export request and fallback plan (§9.5)
Doctor finds it slower than his current EMR	Parity with daily features, templates loaded before launch, shadow days

12.5 Launch plan

1. Close open items (§1.2); start DLT, WhatsApp, Cashfree, Apple, Google, ABDM and DSC paperwork in week 1.
2. Build per the Build Plan; demo to Dr. Gupta every two weeks.
3. Simulated clinic day on staging with dummy data; acceptance criteria pass.
4. Pilot: production go-live, 2-week parallel run with the current EMR, then cut-over.
5. Store launch; patient onboarding at the clinic (ABHA, WhatsApp opt-in, app install); ABDM M3 shortly after.
6. Staff training (one hour each), one-page desk guide, patient FAQ, named support owner with one-business-day response.